Persiapan Dataset dan Validasi Awal

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
!pip install ultralytics
```

```
Collecting ultralytics
  Downloading ultralytics-8.3.240-py3-none-any.whl.metadata (37 kB)
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.2)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (3.10.0)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (4.12.0.88)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.16.3)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.9.0+cu126)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.24.0+cu126)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.31.0)
Collecting ultralytics-thop>=2.0.18 (from ultralytics)
  Downloading ultralytics_thop-2.0.18-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytic
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) ((
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytic
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytic
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytic
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultraly
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ult
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytic
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytic
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.20.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultraly
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (1.14
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (2025
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->u
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->u
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ult
Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ult
Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultra
Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ul
Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->u
Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->u
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ul
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultraly
Requirement already satisfied: nvidia-nvshmem-cu12==3.3.20 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultra
Requirement already satisfied: nvidia-nvtx-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultraly
Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->u
Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ult
Requirement already satisfied: triton==3.5.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.5.
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.8.0
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.8.0->ultralyt
Downloading ultralytics-8.3.240-py3-none-any.whl (1.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.1/1.1 MB 63.7 MB/s eta 0:00:00
Downloading ultralytics_thop-2.0.18-py3-none-any.whl (28 kB)
Installing collected packages: ultralytics-thop, ultralytics
Successfully installed ultralytics-8.3.240 ultralytics-thop-2.0.18
```

```
from ultralytics import YOLO
import ultralytics

print(ultralytics.__version__)
```

```
Creating new Ultralytics Settings v0.0.6 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralytics
8.3.237
```

```python
PROJECT_ROOT = "/content/drive/MyDrive/TA_KopiRobusta_YOLOv8"

import os
os.makedirs(PROJECT_ROOT, exist_ok=True)
```

```python
DATASET_PATH = "/content/drive/MyDrive/TUGAS_AKHIR/Dataset"
```

```python
import os
dataset_path = "/content/drive/MyDrive/TUGAS_AKHIR/Dataset"
os.listdir(dataset_path)
```

```
['data.yaml',
 'README.roboflow.txt',
 'README.dataset.txt',
 'train',
 'valid',
 'test']
```

```python
for split in ["train", "valid", "test"]:
    print(f"\nIsi folder {split}:")
    print(os.listdir(f"{DATASET_PATH}/{split}"))
```

```

Isi folder train:
['labels', 'images']

Isi folder valid:
['images', 'labels']

Isi folder test:
['images', 'labels']
```

```python
with open("/content/drive/MyDrive/TUGAS_AKHIR/Dataset/data.yaml", "r") as f:
    print(f.read())
```

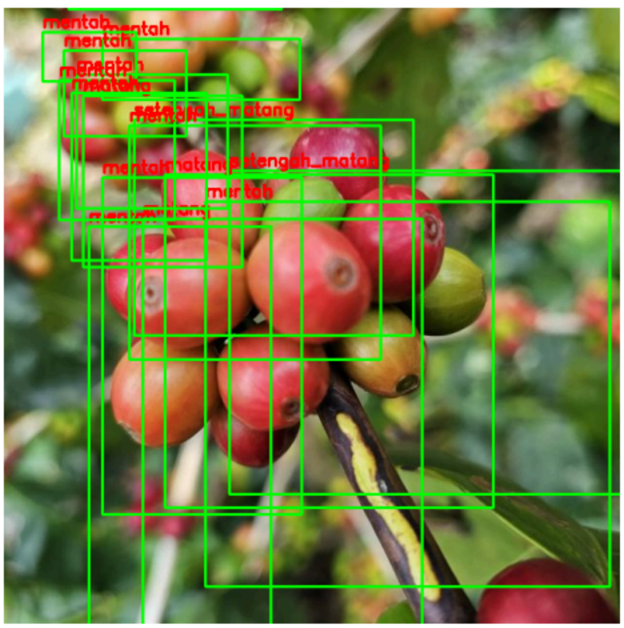
```
train: ../train/images
val: ../valid/images
test: ../test/images

nc: 3
names: ['matang', 'mentah', 'setengah-matang']

roboflow:
  workspace: alfito-dwi-putra
  project: coffee-dataset-2
  version: 5
  license: CC BY 4.0
  url: https://universe.roboflow.com/alfito-dwi-putra/coffee-dataset-2/dataset/5
```

```python
import os
import cv2
import matplotlib.pyplot as plt

# path images dan labels
IMG_DIR = f"{DATASET_PATH}/train/images"
LBL_DIR = f"{DATASET_PATH}/train/labels"

# ambil 1 file contoh
img_file = os.listdir(IMG_DIR)[0]
lbl_file = img_file.replace(".jpg", ".txt").replace(".png", ".txt")

img_path = os.path.join(IMG_DIR, img_file)
lbl_path = os.path.join(LBL_DIR, lbl_file)

img_file, lbl_file
```

```
('20240324_120602_jpg.rf.9285e809187ba7583f5d96d1db857492.jpg',
 '20240324_120602_jpg.rf.9285e809187ba7583f5d96d1db857492.txt')
```

```python
import cv2
import matplotlib.pyplot as plt
import os

IMG_DIR = f"{DATASET_PATH}/train/images"
```

```
LBL_DIR = f"{DATASET_PATH}/train/labels"

img_file = os.listdir(IMG_DIR)[0]
lbl_file = img_file.replace(".jpg", ".txt")

img = cv2.imread(os.path.join(IMG_DIR, img_file))
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
h, w, _ = img.shape

class_names = ["matang", "setengah_matang", "mentah"]

with open(os.path.join(LBL_DIR, lbl_file)) as f:
    for line in f:
        parts = line.split()
        cls, x, y, bw, bh = map(float, parts[:5])

        x1 = int((x - bw/2) * w)
        y1 = int((y - bh/2) * h)
        x2 = int((x + bw/2) * w)
        y2 = int((y + bh/2) * h)

        cv2.rectangle(img, (x1,y1), (x2,y2), (0,255,0), 2)
        cv2.putText(img, class_names[int(cls)],
                    (x1, y1-5),
                    cv2.FONT_HERSHEY_SIMPLEX,
                    0.6, (255,0,0), 2)

plt.figure(figsize=(6,6))
plt.imshow(img)
plt.axis("off")
plt.show()
```



Baseline

```
from ultralytics import YOLO

model = YOLO("yolov8n.pt")

model.train(
    data="/content/drive/MyDrive/TUGAS_AKHIR/Dataset/data.yaml",
    epochs=50,
    imgsz=640,
    batch=16,
    project="/content/drive/MyDrive/TA_KopiRobusta_YOLOv8",
    name="baseline",
    device="0"
)
```

```
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8n.pt to 'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.2MB
Ultralytics 8.3.237 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
engine/trainer: agnostic_nms=False, amp=True, augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7.5, cache=False
Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━ 755.1KB 66.3M
Overriding model.yaml nc=80 with nc=3

                   from  n    params  module                                       arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv             [3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv             [16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f             [32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f             [64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv             [64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f             [128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f             [256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF            [256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f             [384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f             [192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f             [192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f             [384, 256, 1]
 22        [15, 18, 21]  1    751897  ultralytics.nn.modules.head.Detect           [3, [64, 128, 256]]
Model summary: 129 layers, 3,011,433 parameters, 3,011,417 gradients, 8.2 GFLOPs

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt to 'yolo11n.pt': 100% ━━━━━━━━━━━━ 5.4MB
AMP: checks passed ✅
train: Fast image access ✅ (ping: 2.2±3.6 ms, read: 0.1±0.0 MB/s, size: 51.9 KB)
train: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/labels... 1375 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_101214_jpg.rf.d67a2d1937cda830f9a3a6c6914b2b04.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_103600_jpg.rf.b6c050875172ec8ba652610619332734.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_104148_jpg.rf.3cc38dcf3cd1083746ca9d6cf4b33b3c.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120529_jpg.rf.456e92e969dcbe51a15e5b0bb38e9e4d.jpg: 2 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120721_jpg.rf.443ab6eff5f02f614fb6c5ce114f316b.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120727_jpg.rf.f5c6e3378872fd330981e913d01b8631.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_125952_jpg.rf.0c18031256adb7270294011525fcf7b6.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_100729_jpg.rf.e7ea6c2586309b52d631141e8e56cbfc.jpg
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_104324_jpg.rf.240713ce9e6c51ebf613c90da92021b6.jpg
train: New cache created: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/labels.cache
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average
val: Fast image access ✅ (ping: 1.5±1.6 ms, read: 0.1±0.0 MB/s, size: 53.1 KB)
val: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/labels... 394 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_092200_jpg.rf.1547ab2daaa18dc97a505bb3559ab54a.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_110011_jpg.rf.b4691e4df13ccb82557c600f4d27a076.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_125457_jpg.rf.c6086bbc299d662734d7bab246f9629b.jpg: 1 du
val: New cache created: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/labels.cache
Plotting labels to /content/drive/MyDrive/TA_KopiRobusta_YOLOv8/baseline/labels.jpg...
```

```python
# Load model terbaik
best_model_path = "/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/baseline/weights/best.pt"
model = YOLO(best_model_path)

# Evaluasi otomatis
metrics = model.val(
    data="/content/drive/MyDrive/TUGAS_AKHIR/Dataset/data.yaml"
)

# Ambil nilai
precision = metrics.box.mp
recall = metrics.box.mr
map50 = metrics.box.map50
map5095 = metrics.box.map

# Tampilkan
print("=== BASELINE METRICS ===")
print(f"Precision   : {precision:.4f}")
print(f"Recall      : {recall:.4f}")
print(f"mAP50       : {map50:.4f}")
print(f"mAP50-95    : {map5095:.4f}")
```

```
Ultralytics 8.3.237 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
Model summary (fused): 72 layers, 3,006,233 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access ✅ (ping: 0.7±0.6 ms, read: 14.1±8.4 MB/s, size: 51.1 KB)
val: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/labels.cache... 394 images, 0 backgrounds, 0 corrupt: 100% ━━━━
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_092200_jpg.rf.1547ab2daaa18dc97a505bb3559ab54a.jpg: 1 dupl
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_110011_jpg.rf.b4691e4df13ccb82557c600f4d27a076.jpg: 1 dupl
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_125457_jpg.rf.c6086bbc299d662734d7bab246f9629b.jpg: 1 dupl
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 25/25 2.7it/s 9.1s
                   all        394       4155      0.719      0.748      0.807      0.777
                matang        243       1401      0.702      0.572      0.703      0.682
                mentah        270       1389      0.723      0.852      0.868      0.841
        setengah-matang       252       1365      0.734      0.821      0.852      0.808
Speed: 2.7ms preprocess, 4.6ms inference, 0.0ms loss, 2.5ms postprocess per image
Results saved to /content/runs/detect/val
=== BASELINE METRICS ===
Precision   : 0.7194
Recall      : 0.7483
mAP50       : 0.8074
mAP50-95    : 0.7768
```

Learning Rate

```python
from ultralytics import YOLO
import csv

DATA_YAML = "/content/drive/MyDrive/TUGAS_AKHIR/Dataset/data.yaml"
PROJECT_ROOT = "/content/drive/MyDrive/TA_KopiRobusta_YOLOv8"
COARSE_DIR = f"{PROJECT_ROOT}/coarse_lr0"

results = []

for lr in [0.001, 0.01, 0.1]:
    print(f"\n=== Training lr0 = {lr} ===")

    model = YOLO("yolov8n.pt")

    model.train(
        data=DATA_YAML,
        epochs=30,
        imgsz=640,
        batch=16,
        lr0=lr,
        project=COARSE_DIR,
        name=f"lr0_{lr}",
        device=0
    )

    # evaluasi otomatis
    metrics = model.val(data=DATA_YAML)

    results.append([
        lr,
        metrics.box.mp,
        metrics.box.mr,
        metrics.box.map50,
        metrics.box.map
    ])

# simpan hasil
csv_path = f"{COARSE_DIR}/coarse_lr0_results.csv"
with open(csv_path, "w", newline="") as f:
    writer = csv.writer(f)
    writer.writerow(["lr0", "Precision", "Recall", "mAP50", "mAP50-95"])
    writer.writerows(results)

print("Hasil coarse search disimpan di:", csv_path)
```

```
=== Training lr0 = 0.001 ===
Ultralytics 8.3.237 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
engine/trainer: agnostic_nms=False, amp=True, augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7.5, cache=False
Overriding model.yaml nc=80 with nc=3

                   from  n    params  module                                       arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv             [3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv             [16, 32, 3, 2]
```

```
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f             [32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f             [64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv             [64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f             [128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f             [256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF            [256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f             [384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f             [192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f             [192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f             [384, 256, 1]
 22        [15, 18, 21]  1    751897  ultralytics.nn.modules.head.Detect           [3, [64, 128, 256]]
Model summary: 129 layers, 3,011,433 parameters, 3,011,417 gradients, 8.2 GFLOPs

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed ✅
train: Fast image access ✅ (ping: 0.7±0.2 ms, read: 30.6±11.1 MB/s, size: 51.0 KB)
train: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/labels.cache... 1375 images, 0 backgrounds, 0 corrupt: 100% ━
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_101214_jpg.rf.d67a2d1937cda830f9a3a6c6914b2b04.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_103600_jpg.rf.b6c050875172ec8ba652610619332734.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_104148_jpg.rf.3cc38dcf3cd1083746ca9d6cf4b33b3c.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120529_jpg.rf.456e92e969dcbe51a15e5b0bb38e9e4d.jpg: 2 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120721_jpg.rf.443ab6eff5f02f614fb6c5ce114f316b.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120727_jpg.rf.f5c6e3378872fd330981e913d01b8631.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_125952_jpg.rf.0c18031256adb7270294011525fcf7b6.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_100729_jpg.rf.e7ea6c2586309b52d631141e8e56cbfc.jpg
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_104324_jpg.rf.240713ce9e6c51ebf613c90da92021b6.jpg
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average
val: Fast image access ✅ (ping: 0.8±0.7 ms, read: 7.6±4.9 MB/s, size: 52.8 KB)
val: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/labels.cache... 394 images, 0 backgrounds, 0 corrupt: 100% ━━━
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_092200_jpg.rf.1547ab2daaa18dc97a505bb3559ab54a.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_110011_jpg.rf.b4691e4df13ccb82557c600f4d27a076.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_125457_jpg.rf.c6086bbc299d662734d7bab246f9629b.jpg: 1 du
Plotting labels to /content/drive/MyDrive/TA_KopiRobusta_YOLOv8/coarse_lr0/lr0_0.001/labels.jpg...
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.001' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momen
optimizer: AdamW(lr=0.001429, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=
Image sizes 640 train, 640 val
```

Analisis Hasil lr0

```
import pandas as pd

csv_path = "/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/coarse_lr0/coarse_lr0_results.csv"
df = pd.read_csv(csv_path)

df
```

	lr0	Precision	Recall	mAP50	mAP50-95
0	0.001	0.695148	0.794684	0.813687	0.780756
1	0.010	0.695148	0.794684	0.813687	0.780756
2	0.100	0.695148	0.794684	0.813687	0.780756

IMGSZ

```
from ultralytics import YOLO

img_sizes = [320, 640, 800]

for imgsz in img_sizes:
    model = YOLO("yolov8n.pt")

    model.train(
        data="/content/drive/MyDrive/TUGAS_AKHIR/Dataset/data.yaml",
```

```
        epochs=50,
        imgsz=imgsz,
        batch=16,
        project="/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/imgsz",
        name=f"imgsz_{imgsz}",
        device="0"
    )
```

```
Creating new Ultralytics Settings v0.0.6 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralyti
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8n.pt to 'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.2MB
Ultralytics 8.3.239 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
engine/trainer: agnostic_nms=False, amp=True, augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7.5, cache=False
Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━ 755.1KB 28.1MB
Overriding model.yaml nc=80 with nc=3

                   from  n    params  module                                       arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv             [3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv             [16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f             [32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f             [64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv             [64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f             [128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f             [256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF            [256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f             [384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f             [192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f             [192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f             [384, 256, 1]
 22        [15, 18, 21]  1    751897  ultralytics.nn.modules.head.Detect           [3, [64, 128, 256]]
Model summary: 129 layers, 3,011,433 parameters, 3,011,417 gradients, 8.2 GFLOPs

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt to 'yolo11n.pt': 100% ━━━━━━━━━━━━ 5.4MB
AMP: checks passed ✅
train: Fast image access ✅ (ping: 0.7±0.3 ms, read: 0.2±0.1 MB/s, size: 51.9 KB)
train: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/labels.cache... 1375 images, 0 backgrounds, 0 corrupt: 100% ━
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_101214_jpg.rf.d67a2d1937cda830f9a3a6c6914b2b04.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_103600_jpg.rf.b6c050875172ec8ba652610619332734.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_104148_jpg.rf.3cc38dcf3cd1083746ca9d6cf4b33b3c.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120529_jpg.rf.456e92e969dcbe51a15e5b0bb38e9e4d.jpg: 2 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120721_jpg.rf.443ab6eff5f02f614fb6c5ce114f316b.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120727_jpg.rf.f5c6e3378872fd330981e913d01b8631.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_125952_jpg.rf.0c18031256adb7270294011525fcf7b6.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_100729_jpg.rf.e7ea6c2586309b52d631141e8e56cbfc.jpg
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_104324_jpg.rf.240713ce9e6c51ebf613c90da92021b6.jpg
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_averag
val: Fast image access ✅ (ping: 0.6±0.3 ms, read: 0.2±0.0 MB/s, size: 53.1 KB)
val: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/labels.cache... 394 images, 0 backgrounds, 0 corrupt: 100% ━━━━
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_092200_jpg.rf.1547ab2daaa18dc97a505bb3559ab54a.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_110011_jpg.rf.b4691e4df13ccb82557c600f4d27a076.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_125457_jpg.rf.c6086bbc299d662734d7bab246f9629b.jpg: 1 du
```

```
import pandas as pd
import os

BASE_DIR = "/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/imgsz"

imgsz_runs = {
    320: os.path.join(BASE_DIR, "imgsz_320", "results.csv"),
    640: os.path.join(BASE_DIR, "imgsz_640", "results.csv"),
    800: os.path.join(BASE_DIR, "imgsz_800", "results.csv"),
}
```

```
results = []

for imgsz, csv_path in imgsz_runs.items():
```

```
    if not os.path.exists(csv_path):
        print(f"❌ File tidak ditemukan: {csv_path}")
        continue

    df = pd.read_csv(csv_path)

    # ambil epoch terakhir
    last = df.iloc[-1]

    results.append({
        "imgsz": imgsz,
        "Precision": last["metrics/precision(B)"],
        "Recall": last["metrics/recall(B)"],
        "mAP50": last["metrics/mAP50(B)"],
        "mAP50-95": last["metrics/mAP50-95(B)"]
    })
```

```
results_df = pd.DataFrame(results).sort_values("imgsz")
results_df
```

	imgsz	Precision	Recall	mAP50	mAP50-95
0	320	0.71035	0.75171	0.79289	0.73211
1	640	0.71118	0.75073	0.80122	0.76990
2	800	0.68120	0.78055	0.79674	0.77070

BATCH

```
from ultralytics import YOLO

BEST_IMGSZ = 640
batch_sizes = [8, 16, 32]

for batch in batch_sizes:
    model = YOLO("yolov8n.pt")

    model.train(
        data="/content/drive/MyDrive/TUGAS_AKHIR/Dataset/data.yaml",
        epochs=50,
        imgsz=BEST_IMGSZ,
        batch=batch,
        project="/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/batch",
        name=f"batch_{batch}",
        device="0",
        exist_ok=True
    )
```

```
Ultralytics 8.3.239 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
engine/trainer: agnostic_nms=False, amp=True, augment=False, auto_augment=randaugment, batch=8, bgr=0.0, box=7.5, cache=False,
Overriding model.yaml nc=80 with nc=3

                   from  n    params  module                                       arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv             [3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv             [16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f             [32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f             [64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv             [64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f             [128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f             [256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF            [256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f             [384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f             [192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f             [192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f             [384, 256, 1]
```

```
 22        [15, 18, 21]  1    751897  ultralytics.nn.modules.head.Detect           [3, [64, 128, 256]]
Model summary: 129 layers, 3,011,433 parameters, 3,011,417 gradients, 8.2 GFLOPs

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed ✅
train: Fast image access ✅ (ping: 0.5±0.2 ms, read: 25.1±8.9 MB/s, size: 51.0 KB)
train: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/labels.cache... 1375 images, 0 backgrounds, 0 corrupt: 100% ━
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_101214_jpg.rf.d67a2d1937cda830f9a3a6c6914b2b04.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_103600_jpg.rf.b6c050875172ec8ba652610619332734.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_104148_jpg.rf.3cc38dcf3cd1083746ca9d6cf4b33b3c.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120529_jpg.rf.456e92e969dcbe51a15e5b0bb38e9e4d.jpg: 2 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120721_jpg.rf.443ab6eff5f02f614fb6c5ce114f316b.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120727_jpg.rf.f5c6e3378872fd330981e913d01b8631.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_125952_jpg.rf.0c18031256adb7270294011525fcf7b6.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_100729_jpg.rf.e7ea6c2586309b52d631141e8e56cbfc.jpg
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_104324_jpg.rf.240713ce9e6c51ebf613c90da92021b6.jpg
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_averag
val: Fast image access ✅ (ping: 1.8±1.4 ms, read: 12.2±10.6 MB/s, size: 52.8 KB)
val: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/labels.cache... 394 images, 0 backgrounds, 0 corrupt: 100% ━━━━
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_092200_jpg.rf.1547ab2daaa18dc97a505bb3559ab54a.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_110011_jpg.rf.b4691e4df13ccb82557c600f4d27a076.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_125457_jpg.rf.c6086bbc299d662734d7bab246f9629b.jpg: 1 du
Plotting labels to /content/drive/MyDrive/TA_KopiRobusta_YOLOv8/batch/batch_8/labels.jpg...
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momen
optimizer: AdamW(lr=0.001429, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay
Image sizes 640 train, 640 val
Using 2 dataloader workers
Logging results to /content/drive/MyDrive/TA_KopiRobusta_YOLOv8/batch/batch_8
```

```python
import pandas as pd
import os
BASE_DIR = "/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/batch"

batch_runs = {
    8:  os.path.join(BASE_DIR, "batch_8", "results.csv"),
    16: os.path.join(BASE_DIR, "batch_16", "results.csv"),
    32: os.path.join(BASE_DIR, "batch_32", "results.csv"),
}
results = []

for batch, csv_path in batch_runs.items():
    if not os.path.exists(csv_path):
        print(f"❌ File tidak ditemukan: {csv_path}")
        continue

    df = pd.read_csv(csv_path)
    last = df.iloc[-1]

    results.append({
        "batch": batch,
        "Precision": last["metrics/precision(B)"],
        "Recall": last["metrics/recall(B)"],
        "mAP50": last["metrics/mAP50(B)"],
        "mAP50-95": last["metrics/mAP50-95(B)"]
    })
```

```python
results_df = pd.DataFrame(results).sort_values("batch")
results_df
```

	batch	Precision	Recall	mAP50	mAP50-95
0	8	0.70518	0.77909	0.80592	0.77517
1	16	0.71118	0.75073	0.80122	0.76990
2	32	0.71811	0.75294	0.79973	0.76960

```python
best_row = results_df.loc[results_df["mAP50-95"].idxmax()]
best_row
```

	0
batch	8.00000
Precision	0.70518
Recall	0.77909
mAP50	0.80592
mAP50-95	0.77517

dtype: float64

OPTIMIZER

```
from ultralytics import YOLO

IMG_SIZE = 640
BATCH_SIZE = 8
EPOCHS = 50

optimizers = ["SGD", "AdamW"]

for opt in optimizers:
    model = YOLO("yolov8n.pt")

    model.train(
        data="/content/drive/MyDrive/TUGAS_AKHIR/Dataset/data.yaml",
        epochs=EPOCHS,
        imgsz=IMG_SIZE,
        batch=BATCH_SIZE,
        optimizer=opt,
        project="/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/optimizer",
        name=opt,
        device="cpu",
        exist_ok=True
    )
```

```
Creating new Ultralytics Settings v0.0.6 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralyti
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8n.pt to 'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.2MB
Ultralytics 8.3.239 🚀 Python-3.12.12 torch-2.9.0+cpu CPU (Intel Xeon CPU @ 2.20GHz)
engine/trainer: agnostic_nms=False, amp=True, augment=False, auto_augment=randaugment, batch=8, bgr=0.0, box=7.5, cache=False,
Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━ 755.1KB 4.7MB,
Overriding model.yaml nc=80 with nc=3

                   from  n    params  module                                       arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv             [3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv             [16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f             [32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f             [64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv             [64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f             [128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f             [256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF            [256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f             [384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f             [192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f             [192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f             [384, 256, 1]
 22        [15, 18, 21]  1    751897  ultralytics.nn.modules.head.Detect           [3, [64, 128, 256]]
Model summary: 129 layers, 3,011,433 parameters, 3,011,417 gradients, 8.2 GFLOPs

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
train: Fast image access ✅ (ping: 0.7±0.0 ms, read: 0.1±0.0 MB/s, size: 51.9 KB)
train: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/labels.cache... 1375 images, 0 backgrounds, 0 corrupt: 100% ━
```

```
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_101214_jpg.rf.d67a2d1937cda830f9a3a6c6914b2b04.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_103600_jpg.rf.b6c050875172ec8ba652610619332734.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_104148_jpg.rf.3cc38dcf3cd1083746ca9d6cf4b33b3c.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120529_jpg.rf.456e92e969dcbe51a15e5b0bb38e9e4d.jpg: 2 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120721_jpg.rf.443ab6eff5f02f614fb6c5ce114f316b.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120727_jpg.rf.f5c6e3378872fd330981e913d01b8631.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_125952_jpg.rf.0c18031256adb7270294011525fcf7b6.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_100729_jpg.rf.e7ea6c2586309b52d631141e8e56cbfc.jpg
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_104324_jpg.rf.240713ce9e6c51ebf613c90da92021b6.jpg
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average
val: Fast image access ✅ (ping: 0.9±0.2 ms, read: 0.1±0.0 MB/s, size: 53.1 KB)
val: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/labels.cache... 394 images, 0 backgrounds, 0 corrupt: 100% ━━━━
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_092200_jpg.rf.1547ab2daaa18dc97a505bb3559ab54a.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_110011_jpg.rf.b4691e4df13ccb82557c600f4d27a076.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_125457_jpg.rf.c6086bbc299d662734d7bab246f9629b.jpg: 1 du
Plotting labels to /content/drive/MyDrive/TA_KopiRobusta_YOLOv8/optimizer/SGD/labels.jpg...
optimizer: SGD(lr=0.01, momentum=0.937) with parameter groups 57 weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0
Image sizes 640 train, 640 val
```

```python
from ultralytics import YOLO

IMG_SIZE = 640
BATCH_SIZE = 8
EPOCHS = 50

optimizers = ["AdamW"]

for opt in optimizers:
    model = YOLO("yolov8n.pt")

    model.train(
        data="/content/drive/MyDrive/TUGAS_AKHIR/Dataset/data.yaml",
        epochs=EPOCHS,
        imgsz=IMG_SIZE,
        batch=BATCH_SIZE,
        optimizer=opt,
        project="/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/optimizer",
        name=opt,
        device="0",
        exist_ok=True
    )
```

```
Creating new Ultralytics Settings v0.0.6 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralyti
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8n.pt to 'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.2MB
Ultralytics 8.3.239 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
engine/trainer: agnostic_nms=False, amp=True, augment=False, auto_augment=randaugment, batch=8, bgr=0.0, box=7.5, cache=False,
Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━ 755.1KB 26.1M
Overriding model.yaml nc=80 with nc=3

                   from  n    params  module                                       arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv             [3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv             [16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f             [32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f             [64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv             [64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f             [128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f             [256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF            [256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f             [384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f             [192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f             [192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f             [384, 256, 1]
 22        [15, 18, 21]  1    751897  ultralytics.nn.modules.head.Detect           [3, [64, 128, 256]]
Model summary: 129 layers, 3,011,433 parameters, 3,011,417 gradients, 8.2 GFLOPs

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt to 'yolo11n.pt': 100% ━━━━━━━━━━━━ 5.4MB
AMP: checks passed ✅
```

```
train: Fast image access ✅ (ping: 0.5±0.2 ms, read: 0.1±0.0 MB/s, size: 51.9 KB)
train: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/labels.cache... 1375 images, 0 backgrounds, 0 corrupt: 100% ━
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_101214_jpg.rf.d67a2d1937cda830f9a3a6c6914b2b04.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_103600_jpg.rf.b6c050875172ec8ba652610619332734.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_104148_jpg.rf.3cc38dcf3cd1083746ca9d6cf4b33b3c.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120529_jpg.rf.456e92e969dcbe51a15e5b0bb38e9e4d.jpg: 2 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120721_jpg.rf.443ab6eff5f02f614fb6c5ce114f316b.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120727_jpg.rf.f5c6e3378872fd330981e913d01b8631.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_125952_jpg.rf.0c18031256adb7270294011525fcf7b6.jpg: 1 (
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_100729_jpg.rf.e7ea6c2586309b52d631141e8e56cbfc.jpg
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_104324_jpg.rf.240713ce9e6c51ebf613c90da92021b6.jpg
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_averag
val: Fast image access ✅ (ping: 0.5±0.2 ms, read: 0.1±0.0 MB/s, size: 53.1 KB)
val: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/labels.cache... 394 images, 0 backgrounds, 0 corrupt: 100% ━━━━
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_092200_jpg.rf.1547ab2daaa18dc97a505bb3559ab54a.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_110011_jpg.rf.b4691e4df13ccb82557c600f4d27a076.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_125457_jpg.rf.c6086bbc299d662734d7bab246f9629b.jpg: 1 du
```

```
import pandas as pd
import os

BASE_DIR = "/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/optimizer"

opt_runs = {
    "SGD": os.path.join(BASE_DIR, "SGD", "results.csv"),
    "AdamW": os.path.join(BASE_DIR, "AdamW", "results.csv"),
}

results = []

for opt, csv_path in opt_runs.items():
    if not os.path.exists(csv_path):
        print(f"❌ File tidak ditemukan: {csv_path}")
        continue

    df = pd.read_csv(csv_path)
    last = df.iloc[-1]

    results.append({
        "Optimizer": opt,
        "Precision": last["metrics/precision(B)"],
        "Recall": last["metrics/recall(B)"],
        "mAP50": last["metrics/mAP50(B)"],
        "mAP50-95": last["metrics/mAP50-95(B)"]
    })

results_df = pd.DataFrame(results)
results_df
```

	Optimizer	Precision	Recall	mAP50	mAP50-95
0	SGD	0.70773	0.76450	0.80689	0.77076
1	AdamW	0.71869	0.76911	0.82032	0.78729

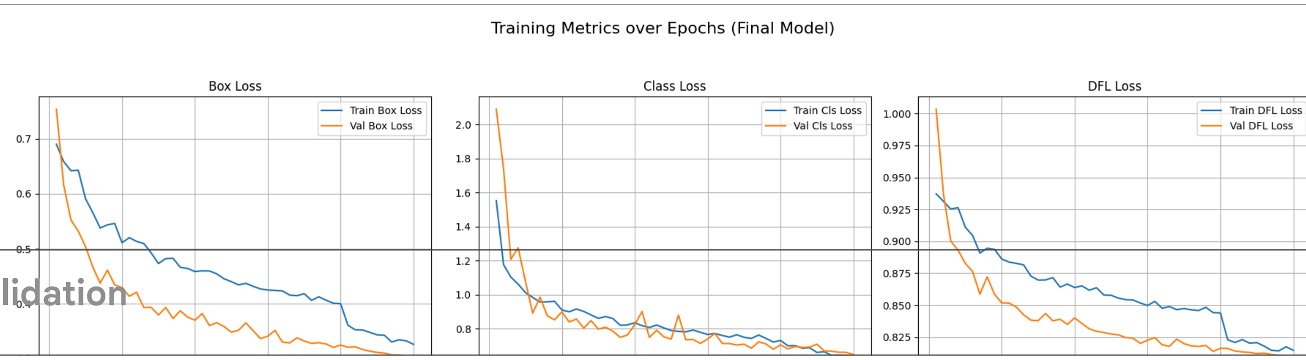
TRAIN FINAL

```
from ultralytics import YOLO

model = YOLO("yolov8n.pt")

model.train(
    data="/content/drive/MyDrive/TUGAS_AKHIR/Dataset/data.yaml",
    epochs=50,
    imgsz=640,
    batch=8,
    optimizer="AdamW",
    project="/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/final_model",
    name="yolov8n_final",
    device="0",
    exist_ok=True
)
```

```
Creating new Ultralytics Settings v0.0.6 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralyti
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8n.pt to 'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.2MB
Ultralytics 8.3.240 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (NVIDIA A100-SXM4-80GB, 81222MiB)
engine/trainer: agnostic_nms=False, amp=True, augment=False, auto_augment=randaugment, batch=8, bgr=0.0, box=7.5, cache=False,
Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━ 755.1KB 103.2
Overriding model.yaml nc=80 with nc=3

                   from  n    params  module                                       arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv             [3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv             [16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f             [32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f             [64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv             [64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f             [128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f             [256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF            [256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f             [384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f             [192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f             [192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f             [384, 256, 1]
 22        [15, 18, 21]  1    751897  ultralytics.nn.modules.head.Detect           [3, [64, 128, 256]]
Model summary: 129 layers, 3,011,433 parameters, 3,011,417 gradients, 8.2 GFLOPs

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt to 'yolo11n.pt': 100% ━━━━━━━━━━━━ 5.4MB
AMP: checks passed ✅
train: Fast image access ✅ (ping: 1.2±1.5 ms, read: 0.1±0.0 MB/s, size: 51.9 KB)
train: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/labels.cache... 1375 images, 0 backgrounds, 0 corrupt: 100% ━
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_101214_jpg.rf.d67a2d1937cda830f9a3a6c6914b2b04.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_103600_jpg.rf.b6c050875172ec8ba652610619332734.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_104148_jpg.rf.3cc38dcf3cd1083746ca9d6cf4b33b3c.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120529_jpg.rf.456e92e969dcbe51a15e5b0bb38e9e4d.jpg: 2 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120721_jpg.rf.443ab6eff5f02f614fb6c5ce114f316b.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_120727_jpg.rf.f5c6e3378872fd330981e913d01b8631.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/20240324_125952_jpg.rf.0c18031256adb7270294011525fcf7b6.jpg: 1 
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_100729_jpg.rf.e7ea6c2586309b52d631141e8e56cbfc.jpg
train: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/train/images/IMG_20240324_104324_jpg.rf.240713ce9e6c51ebf613c90da92021b6.jpg
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_averag
val: Fast image access ✅ (ping: 0.5±0.0 ms, read: 0.1±0.0 MB/s, size: 53.1 KB)
val: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/labels.cache... 394 images, 0 backgrounds, 0 corrupt: 100% ━━━━
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_092200_jpg.rf.1547ab2daaa18dc97a505bb3559ab54a.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_110011_jpg.rf.b4691e4df13ccb82557c600f4d27a076.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_125457_jpg.rf.c6086bbc299d662734d7bab246f9629b.jpg: 1 du
```

```python
import pandas as pd
import matplotlib.pyplot as plt
import os

# Path to the results.csv of the final model
results_path = "/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/final_model/yolov8n_final/results.csv"

if os.path.exists(results_path):
    # Load data
    df = pd.read_csv(results_path)

    # Strip whitespace from column names
    df.columns = df.columns.str.strip()

    # Create subplots
    fig, axs = plt.subplots(2, 3, figsize=(18, 10))
    fig.suptitle('Training Metrics over Epochs (Final Model)', fontsize=16)

    # Plot Losses
    axs[0, 0].plot(df['epoch'], df['train/box_loss'], label='Train Box Loss')
    axs[0, 0].plot(df['epoch'], df['val/box_loss'], label='Val Box Loss')
    axs[0, 0].set_title('Box Loss')
```

```
    axs[0, 0].legend()
    axs[0, 0].grid(True)

    axs[0, 1].plot(df['epoch'], df['train/cls_loss'], label='Train Cls Loss')
    axs[0, 1].plot(df['epoch'], df['val/cls_loss'], label='Val Cls Loss')
    axs[0, 1].set_title('Class Loss')
    axs[0, 1].legend()
    axs[0, 1].grid(True)

    axs[0, 2].plot(df['epoch'], df['train/dfl_loss'], label='Train DFL Loss')
    axs[0, 2].plot(df['epoch'], df['val/dfl_loss'], label='Val DFL Loss')
    axs[0, 2].set_title('DFL Loss')
    axs[0, 2].legend()
    axs[0, 2].grid(True)

    # Plot Metrics
    axs[1, 0].plot(df['epoch'], df['metrics/precision(B)'], label='Precision', color='tab:purple')
    axs[1, 0].plot(df['epoch'], df['metrics/recall(B)'], label='Recall', color='tab:orange')
    axs[1, 0].set_title('Precision & Recall')
    axs[1, 0].legend()
    axs[1, 0].grid(True)

    axs[1, 1].plot(df['epoch'], df['metrics/mAP50(B)'], label='mAP50', color='tab:green')
    axs[1, 1].set_title('mAP @ 0.50')
    axs[1, 1].legend()
    axs[1, 1].grid(True)

    axs[1, 2].plot(df['epoch'], df['metrics/mAP50-95(B)'], label='mAP50-95', color='tab:red')
    axs[1, 2].set_title('mAP @ 0.50:0.95')
    axs[1, 2].legend()
    axs[1, 2].grid(True)

    plt.tight_layout(rect=[0, 0.03, 1, 0.95])
    plt.show()
else:
    print(f"File not found: {results_path}")
```



Validation

```
from ultralytics import YOLO

model = YOLO("/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/final_model/yolov8n_final/weights/best.pt")

model.val(
    data="/content/drive/MyDrive/TUGAS_AKHIR/Dataset/data.yaml",
    imgsz=640,
    batch=8,
    project="/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/evaluation",
    name="validation",
    exist_ok=True
)
```

```
Ultralytics 8.3.240 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (NVIDIA A100-SXM4-80GB, 81222MiB)
Model summary (fused): 72 layers, 3,006,233 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access ✅ (ping: 1.0±1.5 ms, read: 22.5±13.1 MB/s, size: 53.1 KB)
val: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/labels.cache... 394 images, 0 backgrounds, 0 corrupt: 100% ━━━
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_092200_jpg.rf.1547ab2daaa18dc97a505bb3559ab54a.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_110011_jpg.rf.b4691e4df13ccb82557c600f4d27a076.jpg: 1 du
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/valid/images/20240324_125457_jpg.rf.c6086bbc299d662734d7bab246f9629b.jpg: 1 du
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 50/50 12.9it/s 3.9
                   all        394       4155      0.713      0.781      0.823       0.79
                matang        243       1401      0.664      0.685      0.731      0.709
                mentah        270       1389      0.742      0.841      0.876      0.844
       setengah-matang        252       1365      0.734      0.819      0.861      0.816
Speed: 1.1ms preprocess, 1.7ms inference, 0.0ms loss, 0.9ms postprocess per image
```

```
Results saved to /content/drive/MyDrive/TA_KopiRobusta_YOLOv8/evaluation/validation
ultralytics.utils.metrics.DetMetrics object with attributes:

ap_class_index: array([0, 1, 2])
box: ultralytics.utils.metrics.Metric object
confusion_matrix: <ultralytics.utils.metrics.ConfusionMatrix object at 0x7c03e3ca4680>
curves: ['Precision-Recall(B)', 'F1-Confidence(B)', 'Precision-Confidence(B)', 'Recall-Confidence(B)']
curves_results: [[array([          0,    0.001001,    0.002002,    0.003003,    0.004004,    0.005005,    0.006006,
0.007007,    0.008008,    0.009009,     0.01001,    0.011011,    0.012012,    0.013013,    0.014014,    0.015015,    0.016016,
0.017017,    0.018018,    0.019019,     0.02002,    0.021021,    0.022022,    0.023023,
          0.024024,    0.025025,    0.026026,    0.027027,    0.028028,    0.029029,     0.03003,    0.031031,    0.032032,
0.033033,    0.034034,    0.035035,    0.036036,    0.037037,    0.038038,    0.039039,     0.04004,    0.041041,    0.042042,
0.043043,    0.044044,    0.045045,    0.046046,    0.047047,
          0.048048,    0.049049,     0.05005,    0.051051,    0.052052,    0.053053,    0.054054,    0.055055,    0.056056,
0.057057,    0.058058,    0.059059,     0.06006,    0.061061,    0.062062,    0.063063,    0.064064,    0.065065,    0.066066,
0.067067,    0.068068,    0.069069,     0.07007,    0.071071,
          0.072072,    0.073073,    0.074074,    0.075075,    0.076076,    0.077077,    0.078078,    0.079079,     0.08008,
0.081081,    0.082082,    0.083083,    0.084084,    0.085085,    0.086086,    0.087087,    0.088088,    0.089089,     0.09009,
0.091091,    0.092092,    0.093093,    0.094094,    0.095095,
          0.096096,    0.097097,    0.098098,    0.099099,      0.1001,      0.1011,      0.1021,      0.1031,      0.1041,
0.10511,     0.10611,     0.10711,     0.10811,     0.10911,     0.11011,     0.11111,     0.11211,     0.11311,     0.11411,
0.11512,     0.11612,     0.11712,     0.11812,     0.11912,
           0.12012,     0.12112,     0.12212,     0.12312,     0.12412,     0.12513,     0.12613,     0.12713,     0.12813,
0.12913,     0.13013,     0.13113,     0.13213,     0.13313,     0.13413,     0.13514,     0.13614,     0.13714,     0.13814,
0.13914,     0.14014,     0.14114,     0.14214,     0.14314,
           0.14414,     0.14515,     0.14615,     0.14715,     0.14815,     0.14915,     0.15015,     0.15115,     0.15215,
0.15315,     0.15415,     0.15516,     0.15616,     0.15716,     0.15816,     0.15916,     0.16016,     0.16116,     0.16216,
0.16316,     0.16416,     0.16517,     0.16617,     0.16717,
           0.16817,     0.16917,     0.17017,     0.17117,     0.17217,     0.17317,     0.17417,     0.17518,     0.17618,
0.17718,     0.17818,     0.17918,     0.18018,     0.18118,     0.18218,     0.18318,     0.18418,     0.18519,     0.18619,
0.18719,     0.18819,     0.18919,     0.19019,     0.19119,
           0.19219,     0.19319,     0.19419,      0.1952,      0.1962,      0.1972,      0.1982,      0.1992,      0.2002,
0.2012,      0.2022,      0.2032,      0.2042,     0.20521,     0.20621,     0.20721,     0.20821,     0.20921,     0.21021,
0.21121,     0.21221,     0.21321,     0.21421,     0.21522,
           0.21622,     0.21722,     0.21822,     0.21922,     0.22022,     0.22122,     0.22222,     0.22322,     0.22422,
0.22523,     0.22623,     0.22723,     0.22823,     0.22923,     0.23023,     0.23123,     0.23223,     0.23323,     0.23423,
0.23524,     0.23624,     0.23724,     0.23824,     0.23924,
           0.24024,     0.24124,     0.24224,     0.24324,     0.24424,     0.24525,     0.24625,     0.24725,     0.24825,
0.24925,     0.25025,     0.25125,     0.25225,     0.25325,     0.25425,     0.25526,     0.25626,     0.25726,     0.25826,
0.25926,     0.26026,     0.26126,     0.26226,     0.26326,
           0.26426,     0.26527,     0.26627,     0.26727,     0.26827,     0.26927,     0.27027,     0.27127,     0.27227,
0.27327,     0.27427,     0.27528,     0.27628,     0.27728,     0.27828,     0.27928,     0.28028,     0.28128,     0.28228,
0.28328,     0.28428,     0.28529,     0.28629,     0.28729,
```

TESTING

```
model.val(
    data="/content/drive/MyDrive/TUGAS_AKHIR/Dataset/data.yaml",
    split="test",
    imgsz=640,
    batch=8,
    project="/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/evaluation",
    name="test",
    exist_ok=True
)
```

```
Ultralytics 8.3.240 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (NVIDIA A100-SXM4-80GB, 81222MiB)
val: Fast image access ✅ (ping: 1.0±0.9 ms, read: 12.8±11.5 MB/s, size: 48.5 KB)
val: Scanning /content/drive/MyDrive/TUGAS_AKHIR/Dataset/test/labels.cache... 196 images, 0 backgrounds, 0 corrupt: 100% ━━━━
val: /content/drive/MyDrive/TUGAS_AKHIR/Dataset/test/images/20240324_090819_jpg.rf.ca5cbeb40a306f75a1cec535b0652acd.jpg: 1 dup
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 25/25 8.8it/s 2.8s
                   all        196       2104      0.723      0.781      0.825      0.793
                matang        131        740      0.675      0.703      0.733      0.713
                mentah        134        648      0.751      0.821      0.873      0.844
       setengah-matang        132        716      0.743      0.819       0.87      0.822
Speed: 1.3ms preprocess, 2.8ms inference, 0.0ms loss, 1.3ms postprocess per image
Results saved to /content/drive/MyDrive/TA_KopiRobusta_YOLOv8/evaluation/test
ultralytics.utils.metrics.DetMetrics object with attributes:

ap_class_index: array([0, 1, 2])
box: ultralytics.utils.metrics.Metric object
confusion_matrix: <ultralytics.utils.metrics.ConfusionMatrix object at 0x7c03fa5f90d0>
curves: ['Precision-Recall(B)', 'F1-Confidence(B)', 'Precision-Confidence(B)', 'Recall-Confidence(B)']
curves_results: [[array([          0,    0.001001,    0.002002,    0.003003,    0.004004,    0.005005,    0.006006,
0.007007,    0.008008,    0.009009,     0.01001,    0.011011,    0.012012,    0.013013,    0.014014,    0.015015,    0.016016,
0.017017,    0.018018,    0.019019,     0.02002,    0.021021,    0.022022,    0.023023,
          0.024024,    0.025025,    0.026026,    0.027027,    0.028028,    0.029029,     0.03003,    0.031031,    0.032032,
0.033033,    0.034034,    0.035035,    0.036036,    0.037037,    0.038038,    0.039039,     0.04004,    0.041041,    0.042042,
0.043043,    0.044044,    0.045045,    0.046046,    0.047047,
          0.048048,    0.049049,     0.05005,    0.051051,    0.052052,    0.053053,    0.054054,    0.055055,    0.056056,
```

```
0.057057,   0.058058,   0.059059,    0.06006,   0.061061,   0.062062,   0.063063,   0.064064,   0.065065,   0.066066,
0.067067,   0.068068,   0.069069,    0.07007,   0.071071,
          0.072072,   0.073073,   0.074074,   0.075075,   0.076076,   0.077077,   0.078078,   0.079079,    0.08008,
0.081081,   0.082082,   0.083083,   0.084084,   0.085085,   0.086086,   0.087087,   0.088088,   0.089089,    0.09009,
0.091091,   0.092092,   0.093093,   0.094094,   0.095095,
          0.096096,   0.097097,   0.098098,   0.099099,     0.1001,     0.1011,     0.1021,     0.1031,     0.1041,
0.10511,    0.10611,    0.10711,    0.10811,    0.10911,    0.11011,    0.11111,    0.11211,    0.11311,    0.11411,
0.11512,    0.11612,    0.11712,    0.11812,    0.11912,
           0.12012,    0.12112,    0.12212,    0.12312,    0.12412,    0.12513,    0.12613,    0.12713,    0.12813,
0.12913,    0.13013,    0.13113,    0.13213,    0.13313,    0.13413,    0.13514,    0.13614,    0.13714,    0.13814,
0.13914,    0.14014,    0.14114,    0.14214,    0.14314,
           0.14414,    0.14515,    0.14615,    0.14715,    0.14815,    0.14915,    0.15015,    0.15115,    0.15215,
0.15315,    0.15415,    0.15516,    0.15616,    0.15716,    0.15816,    0.15916,    0.16016,    0.16116,    0.16216,
0.16316,    0.16416,    0.16517,    0.16617,    0.16717,
           0.16817,    0.16917,    0.17017,    0.17117,    0.17217,    0.17317,    0.17417,    0.17518,    0.17618,
0.17718,    0.17818,    0.17918,    0.18018,    0.18118,    0.18218,    0.18318,    0.18418,    0.18519,    0.18619,
0.18719,    0.18819,    0.18919,    0.19019,    0.19119,
           0.19219,    0.19319,    0.19419,     0.1952,     0.1962,     0.1972,     0.1982,     0.1992,     0.2002,
0.2012,     0.2022,     0.2032,     0.2042,    0.20521,    0.20621,    0.20721,    0.20821,    0.20921,    0.21021,
0.21121,    0.21221,    0.21321,    0.21421,    0.21522,
           0.21622,    0.21722,    0.21822,    0.21922,    0.22022,    0.22122,    0.22222,    0.22322,    0.22422,
0.22523,    0.22623,    0.22723,    0.22823,    0.22923,    0.23023,    0.23123,    0.23223,    0.23323,    0.23423,
0.23524,    0.23624,    0.23724,    0.23824,    0.23924,
           0.24024,    0.24124,    0.24224,    0.24324,    0.24424,    0.24525,    0.24625,    0.24725,    0.24825,
0.24925,    0.25025,    0.25125,    0.25225,    0.25325,    0.25425,    0.25526,    0.25626,    0.25726,    0.25826,
0.25926,    0.26026,    0.26126,    0.26226,    0.26326,
           0.26426,    0.26527,    0.26627,    0.26727,    0.26827,    0.26927,    0.27027,    0.27127,    0.27227,
0.27327,    0.27427,    0.27528,    0.27628,    0.27728,    0.27828,    0.27928,    0.28028,    0.28128,    0.28228,
0.28328,    0.28428,    0.28529,    0.28629,    0.28729,
           0.28829,    0.28929,    0.29029,    0.29129,    0.29229,    0.29329,    0.29429,     0.2953,     0.2963,
0.2973,     0.2983,     0.2993,     0.3003,     0.3013,     0.3023,     0.3033,     0.3043,    0.30531,    0.30631,
0.30731,    0.30831,    0.30931,    0.31031,    0.31131,
           0.31231,    0.31331,    0.31431,    0.31532,    0.31632,    0.31732,    0.31832,    0.31932,    0.32032,
```

REAL CASE

```python
from ultralytics import YOLO

model = YOLO("/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/final_model/yolov8n_final/weights/best.pt")

model.predict(
    source="/content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/",
    imgsz=640,
    conf=0.25,
    save=True,
    project="/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/real_case",
    name="inference",
    exist_ok=True
)
```

```
image 1/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/1.jpg: 640x384 2 setengah-matangs, 142.4ms
image 2/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/10.jpg: 640x384 2 setengah-matangs, 7.9ms
image 3/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/11.jpg: 640x384 13 matangs, 2 mentahs, 5 setengah-matangs,
image 4/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/12.jpg: 640x384 7 matangs, 2 mentahs, 3 setengah-matangs, 7
image 5/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/13.jpg: 640x384 (no detections), 7.8ms
image 6/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/14.jpg: 640x384 51 matangs, 6 mentahs, 49 setengah-matangs,
image 7/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/15.jpg: 640x384 37 matangs, 5 mentahs, 15 setengah-matangs,
image 8/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/16.jpg: 640x384 33 matangs, 2 mentahs, 6 setengah-matangs,
image 9/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/17.JPG: 640x384 1 matang, 8.1ms
image 10/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/18.JPG: 640x448 4 mentahs, 13 setengah-matangs, 67.7ms
image 11/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/19.JPG: 640x448 1 matang, 13 setengah-matangs, 7.9ms
image 12/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/2.jpg: 640x384 13 matangs, 1 mentah, 5 setengah-matangs, 8
image 13/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/20.JPG: 640x448 4 matangs, 1 mentah, 19 setengah-matangs,
image 14/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/21.JPG: 640x448 3 matangs, 15 mentahs, 1 setengah-matang,
image 15/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/22.JPG: 640x448 31 mentahs, 7 setengah-matangs, 7.9ms
image 16/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/23.JPG: 640x448 25 mentahs, 2 setengah-matangs, 7.8ms
image 17/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/24.JPG: 640x448 6 matangs, 1 mentah, 8 setengah-matangs, 8
image 18/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/25.jpg: 640x384 6 matangs, 12 setengah-matangs, 8.5ms
image 19/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/3.jpg: 640x384 5 matangs, 6 setengah-matangs, 7.8ms
image 20/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/4.jpg: 640x384 (no detections), 8.4ms
image 21/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/5.jpg: 640x384 5 matangs, 4 mentahs, 18 setengah-matangs,
image 22/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/6.jpg: 640x384 5 matangs, 6 setengah-matangs, 7.8ms
image 23/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/7.jpg: 640x384 7 matangs, 2 mentahs, 22 setengah-matangs,
image 24/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/8.jpg: 640x384 5 matangs, 1 mentah, 8 setengah-matangs, 7.
image 25/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/9.jpg: 640x384 2 matangs, 7.9ms
Speed: 3.1ms preprocess, 15.8ms inference, 1.5ms postprocess per image at shape (1, 3, 640, 384)
Results saved to /content/drive/MyDrive/TA_KopiRobusta_YOLOv8/real_case/inference
[ultralytics.engine.results.Results object with attributes:
```

```
 boxes: ultralytics.engine.results.Boxes object
 keypoints: None
 masks: None
 names: {0: 'matang', 1: 'mentah', 2: 'setengah-matang'}
 obb: None
 orig_img: array([[[216, 239, 217],
         [195, 221, 198],
         [178, 206, 183],
         ...,
         [115, 108,  89],
         [119, 112,  93],
         [123, 116,  97]],

        [[210, 233, 211],
         [187, 213, 190],
         [169, 197, 174],
         ...,
         [115, 108,  89],
         [119, 112,  93],
         [123, 116,  97]],

        [[202, 224, 200],
         [177, 201, 177],
         [156, 182, 158],
         ...,
         [115, 108,  89],
         [119, 112,  93],
```

```python
from ultralytics import YOLO

model = YOLO("/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/final_model/yolov8n_final/weights/best.pt")

model.predict(
    source="/content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/",
    imgsz=640,
    conf=0.25,
    save=True,
    save_txt=True,
    project="/content/drive/MyDrive/TA_KopiRobusta_YOLOv8/real_case",
    name="inference_graph",
    exist_ok=True
)
```

```
image 1/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/1.jpg: 640x384 2 setengah-matangs, 10.5ms
image 2/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/10.jpg: 640x384 2 setengah-matangs, 7.9ms
image 3/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/11.jpg: 640x384 13 matangs, 2 mentahs, 5 setengah-matangs,
image 4/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/12.jpg: 640x384 7 matangs, 2 mentahs, 3 setengah-matangs, 7
image 5/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/13.jpg: 640x384 (no detections), 8.2ms
image 6/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/14.jpg: 640x384 51 matangs, 6 mentahs, 49 setengah-matangs,
image 7/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/15.jpg: 640x384 37 matangs, 5 mentahs, 15 setengah-matangs,
image 8/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/16.jpg: 640x384 33 matangs, 2 mentahs, 6 setengah-matangs,
image 9/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/17.JPG: 640x384 1 matang, 8.0ms
image 10/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/18.JPG: 640x448 4 mentahs, 13 setengah-matangs, 8.6ms
image 11/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/19.JPG: 640x448 1 matang, 13 setengah-matangs, 7.8ms
image 12/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/2.jpg: 640x384 13 matangs, 1 mentah, 5 setengah-matangs, 8
image 13/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/20.JPG: 640x448 4 matangs, 1 mentah, 19 setengah-matangs,
image 14/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/21.JPG: 640x448 3 matangs, 15 mentahs, 1 setengah-matang,
image 15/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/22.JPG: 640x448 31 mentahs, 7 setengah-matangs, 8.0ms
image 16/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/23.JPG: 640x448 25 mentahs, 2 setengah-matangs, 7.9ms
image 17/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/24.JPG: 640x448 6 matangs, 1 mentah, 8 setengah-matangs, 7
image 18/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/25.jpg: 640x384 6 matangs, 12 setengah-matangs, 8.9ms
image 19/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/3.jpg: 640x384 5 matangs, 6 setengah-matangs, 7.9ms
image 20/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/4.jpg: 640x384 (no detections), 8.7ms
image 21/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/5.jpg: 640x384 5 matangs, 4 mentahs, 18 setengah-matangs,
image 22/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/6.jpg: 640x384 5 matangs, 6 setengah-matangs, 7.8ms
image 23/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/7.jpg: 640x384 7 matangs, 2 mentahs, 22 setengah-matangs,
image 24/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/8.jpg: 640x384 5 matangs, 1 mentah, 8 setengah-matangs, 7.
image 25/25 /content/drive/MyDrive/TUGAS_AKHIR/Dataset_Robusta_Ijen/9.jpg: 640x384 2 matangs, 10.0ms
Speed: 3.2ms preprocess, 8.2ms inference, 1.5ms postprocess per image at shape (1, 3, 640, 384)
Results saved to /content/drive/MyDrive/TA_KopiRobusta_YOLOv8/real_case/inference_graph
23 labels saved to /content/drive/MyDrive/TA_KopiRobusta_YOLOv8/real_case/inference_graph/labels
[ultralytics.engine.results.Results object with attributes:

 boxes: ultralytics.engine.results.Boxes object
 keypoints: None
 masks: None
 names: {0: 'matang', 1: 'mentah', 2: 'setengah-matang'}
 obb: None
 orig_img: array([[[216, 239, 217],
         [195, 221, 198],
         [178, 206, 183],
```

```
        ...,
        [115, 108,  89],
        [119, 112,  93],
        [123, 116,  97]],

       [[210, 233, 211],
        [187, 213, 190],
        [169, 197, 174],
        ...,
        [115, 108,  89],
        [119, 112,  93],
        [123, 116,  97]],

       [[202, 224, 200],
        [177, 201, 177],
        [156, 182, 158],
        ...,
        [115, 108,  89],
```